

RUSTENBURG, South Africa

WMO: 682550

Lat: 25.6608S

Lon: 27.2322E

Elev: 3773

StdP: 12.80

Time Zone: 2.00 (E02)

Period: 98-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	36.0	38.9	9.5	10.3	65.0	13.8	12.7	65.1	12.0	60.3	10.0	60.2	2.5	230	0.297

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	19.6	93.9	62.9	91.5	63.2	89.3	62.8	72.7	82.7	71.5	81.2	70.4	79.5	5.7	300

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
70.1	127.4	76.7	68.9	122.2	75.6	67.7	117.1	74.5	39.2	82.6	38.0	81.4	36.9	80.0	85.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 13.1	11.4	9.6	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(5)			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	67.3	75.0	74.9	72.1	66.2	60.6	55.9	55.0	60.7	67.9	71.8	73.1	74.6		
	DBStd	8.50	4.33	3.87	4.28	4.47	4.46	3.51	3.43	5.06	5.59	5.31	4.97	4.71		
	HDD50	10	0	0	0	0	2	3	4	1	0	0	0	0		
	HDD65	974	2	0	5	36	146	273	309	151	37	9	5	1		
	CDD50	6311	774	696	685	485	329	180	161	332	537	676	692	764		
	CDD65	1800	311	276	225	71	8	0	0	17	124	220	248	300		
	CDH74	15763	2537	2107	1461	519	228	20	20	395	1553	2213	2164	2546		
	CDH80	5650	952	748	417	92	18	0	0	66	599	905	846	1007		
(14) Wind	WSAvg	3.3	3.2	3.0	3.9	3.9	3.1	2.6	2.9	3.4	3.6	3.6	3.7	3.1		
(15) Precipitation	PrecAvg	24.2	4.3	3.4	3.3	1.7	0.7	0.2	0.1	0.2	0.6	2.2	3.1	4.1		
	PrecMax	42.4	9.1	12.5	11.1	5.0	4.4	1.7	0.6	1.3	2.2	5.7	6.2	7.2		
	PrecMin	14.6	1.1	1.3	0.3	0.2	0.0	0.0	0.0	0.0	0.0	0.3	0.9	1.8		
	PrecStd	6.5	2.0	2.5	2.3	1.3	1.0	0.4	0.2	0.3	0.7	1.4	1.4	1.5		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	95.7	94.7	91.5	87.3	82.6	76.7	76.7	85.2	92.5	95.4	95.7	97.1		
		MCWB	65.6	64.5	66.2	66.0	55.6	53.1	51.7	56.7	58.0	60.1	61.3	63.6		
	2%	DB	91.3	91.2	87.6	82.6	79.0	73.6	73.2	81.8	89.5	91.8	91.8	92.6		
		MCWB	66.9	64.5	66.0	62.9	56.0	52.3	50.7	55.6	57.2	60.3	61.8	64.5		
	5%	DB	88.3	87.7	84.5	79.9	76.6	71.3	70.2	78.7	87.0	88.6	88.5	88.9		
		MCWB	66.6	65.6	65.9	62.3	56.0	52.0	50.5	53.7	56.3	60.1	62.5	65.3		
	10%	DB	85.6	84.5	81.8	77.1	73.4	68.3	67.6	75.1	83.4	85.3	85.2	85.3		
		MCWB	66.5	65.5	65.6	62.4	55.9	51.4	50.0	52.6	55.8	59.8	63.0	65.4		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	73.9	73.4	73.6	70.1	63.4	59.1	56.3	59.9	64.3	69.2	72.2	74.2		
		MCDB	85.5	85.2	83.7	78.5	74.1	70.4	68.1	78.5	79.2	77.5	81.8	82.4		
	2%	WB	72.4	71.2	71.0	68.1	61.0	56.2	54.4	58.1	61.6	67.0	70.6	72.4		
		MCDB	83.8	80.2	81.0	77.4	71.0	66.8	66.1	76.7	79.5	77.1	79.6	80.9		
	5%	WB	71.1	69.7	69.3	66.4	59.3	54.4	52.7	56.6	59.7	65.3	68.9	71.0		
		MCDB	81.7	77.9	78.9	75.2	70.2	65.5	64.9	74.7	77.6	75.8	77.9	79.5		
	10%	WB	69.7	68.4	67.9	64.8	57.6	52.8	51.0	54.8	58.2	63.6	67.2	69.6		
		MCDB	79.0	76.6	77.3	72.9	68.5	64.4	64.1	71.8	76.4	75.1	76.8	77.9		
(35) Mean Daily Temperature Range	5% DB	MDBR	19.6	19.0	19.1	22.1	27.1	27.7	27.5	28.7	28.7	24.5	21.2	19.8		
		MCDBR	25.0	24.8	23.7	26.4	30.5	31.2	30.9	32.1	33.3	29.9	27.0	25.4		
		MCWBR	7.6	8.3	8.3	10.1	12.3	13.7	13.3	13.1	11.5	9.3	8.1	7.5		
	5% WB	MCDBR	21.5	18.4	19.7	19.9	24.9	26.5	26.0	29.1	27.3	21.6	20.0	19.4		
		MCWBR	8.8	8.1	9.1	9.3	12.5	13.7	13.0	13.4	11.2	8.9	9.1	9.1		
(40) Clear-Sky Solar Irradiance	taub		0.395	0.399	0.387	0.343	0.303	0.287	0.288	0.347	0.390	0.421	0.376	0.382		
	taud		2.345	2.327	2.342	2.427	2.507	2.527	2.503	2.341	2.237	2.192	2.363	2.372		
	Ebn at Noon		301	295	289	288	287	286	290	282	283	285	305	305		
	Edh at Noon		43	43	40	33	28	26	28	36	43	48	42	42		
(44) All-Sky Solar Radiation	RadAvg		2121	2002	1781	1523	1401	1269	1378	1630	1943	2092	2165	2175		
	RadStd		225	187	170	140	66	66	36	76	82	136	205	138		

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47) Regional (13 neighbors)		+1.19	+1.04	N/A	N/A	N/A	N/A	N/A	-326	+421	+157	

Nomenclature: See separate page